Correcting Compound Mass and Yield for Residual Solvent Using ^{1}H NMR

When an isolated product contains residual solvent (e.g., ethyl acetate), the measured mass reflects the total mixture rather than the pure target compound. Quantitative analysis of the ^{1}H NMR spectrum allows you to determine the molar ratio of the components, calculate the true mass of your product, and determine an accurate reaction yield.

Note: this calculation provides a solid estimation for routine lab reports. For rigorous quantitative NMR (qNMR), acquisition parameters—especially the relaxation delay must be adjusted to prevent signal saturation.

1. General Principles & Formulae

Select one well-resolved, non-overlapping resonance from your target compound and one from the residual solvent. Identify the raw integral (I) and the number of contributing protons (n_{H}) for each signal.

Step 1: Normalize Integral Values per Proton

Convert raw peak integrals (I) into normalized molar values (N) by dividing by the number of contributing nuclei (n_{H}):

$$N_{\text{compound}} = \frac{I_{\text{compound}}}{n_{\text{H, compound}}}$$

$$N_{\text{solvent}} = \frac{I_{\text{solvent}}}{n_{\text{H, solvent}}}$$

Step 2: Calculate Relative Mass Contributions

Multiply each normalized molar unit by the corresponding molecular weight (MW):

$$m'_{\text{compound}} = N_{\text{compound}} \times \text{MW}_{\text{compound}}$$

$$m'_{\text{solvent}} = N_{\text{solvent}} \times \text{MW}_{\text{solvent}}$$

Step 3: Determine the Weight Percentage (wt%)

Determine the mass fraction of the target compound relative to the total mixture:

$$\text{wt\%}_{\text{compound}} = \frac{m'_{\text{compound}}}{m'_{\text{compound}} + m'_{\text{solvent}}} \times 100\%$$

Step 4: Calculate Corrected Mass and Reaction Yield

Multiply the weight fraction by the experimentally measured total mass (m_{total}):

$$m_{\text{pure compound}} = m_{\text{total}} \times \frac{\text{wt\%}_{\text{compound}}}{100\%}$$

$$\text{Corrected Yield (\%)} = \frac{m_{\text{pure compound}}}{\text{Theoretical Yield}} \times 100\%$$

2. Worked Example: Benzil with Residual Ethyl Acetate

Assumed sample mass isolated: $m_{\text{total}} = 0.520\ \text{g}$

Experimental Data:

Component	Resonance Chosen	Number of Protons (nH)	Integration (I)	Molecular Weight (MW)
Benzil ($C_{14}H_{10}O_2$)	Ortho protons (*o*-H)	4	4.00	210.23 g/mol
Ethyl Acetate ($C_4H_8O_2$)	$-CH_2-$ quartet (~ 4.12 ppm)	2	0.65	88.11 g/mol

Step-by-Step Calculation

1. **Normalized Molar Values (N):**

$$N_{\text{benzil}} = \frac{4.00}{4} = 1.000$$

$$N_{\text{EtOAc}} = \frac{0.65}{2} = 0.325$$

(Molar ratio = 1.000 mol Benzil: 0.325 mol EtOAc*)*

2. **Relative Mass Contributions (m'):**

$$m'_{\text{benzil}} = 1.000 \times 210.23\ \text{g/mol} = 210.23\ \text{g}$$

$$m'_{\text{EtOAc}} = 0.325 \times 88.11\ \text{g/mol} = 28.64\ \text{g}$$

$$m'_{\text{total}} = 210.23\ \text{g} + 28.64\ \text{g} = 238.87\ \text{g}$$

3. **Weight Percentages (wt%):**

$$\text{wt\%}_{\text{benzil}} = \frac{210.23}{238.87} \times 100\% \approx 88.0\%$$

$$\text{wt\%}_{\text{EtOAc}} = \frac{28.64}{238.87} \times 100\% \approx 12.0\%$$

4. **Corrected Mass and Yield:**

$$m_{\text{pure benzil}} = 0.520\ \text{g} \times 0.880 = 0.458\ \text{g}$$